

Education

- 2022 - 2025 **PhD in Computer Science and Engineering** at University of Bologna – Excellent.
Thesis: *Neural Understanding of Objects and Materials.*
- 2020 - 2022 **M.Sc. in Artificial Intelligence** at University of Bologna – 110/110, with honours.
Thesis: *Mitigating non-Lambertian surfaces issues in Stereo Matching with Neural Radiance Fields.*
- 2017 - 2020 **B.Sc. in Automation Engineering** at University of Bologna – 104/110.
Thesis: *Machine learning. Theoretical principles and applications.*
- 2012 - 2017 **High School Diploma in Automation** at IIS A. Volta – 100/100.
Winner of the XXXI Edition of the Lucchese Prize.

Relevant Work Experience

- since 2025 **Postdoctoral Researcher** at Computer Vision Lab, University of Bologna.
- 2024 - 2025 **Visiting Researcher** at Torr Vision Group, University of Oxford.

Teaching Activities

- since 2022 **Teaching Assistant** at University of Bologna. *Computer Vision and Image Processing* (M.Sc.).
- since 2024 **Teaching Assistant** at University of Bologna. *Image Processing and Computer Vision* (M.Sc.).

Supervision Activities

- 2025 Leonardo Montrucchio. *Learning to be a Multimodal LLM: Investigating Visual-Textual Inconsistency for Industrial Anomaly Detection*, University of Bologna, M.Sc. in Statistical Sciences. [LM-DM270]
- 2025 Lucia Gasperini. *A real multi-view dataset for multimodal Anomaly Detection and Segmentation*, University of Bologna, M.Sc. in Artificial Intelligence. [LM-DM270]
- 2025 Luigi Lella. *A synthetic multi-view dataset for multimodal Anomaly Detection and Segmentation*, University of Bologna, M.Sc. in Computer Engineering. [LM-DM270]
- 2024 Agostino Aiezzo. *Test Time Training for Binary Anomaly Segmentation*, University of Bologna, M.Sc. in Artificial Intelligence. [LM-DM270]
- 2024 Mirko Del Moro. *A Test Time Adaptation Protocol to improve Industrial Anomaly Detection and Segmentation*, University of Bologna, M.Sc. in Artificial Intelligence. [LM-DM270]
- 2023 Sofia Caltabiano. *Creation of a dataset for Instance Segmentation of Transparent and Mirror surfaces*, University of Bologna, Internship in Automation Engineering. [LM-DM270]
- 2023 Andrea Lo Russo. *Monocular Neural Networks for Depth Estimation of non-Lambertian Surfaces*, University of Bologna, M.Sc. in Computer Engineering. [LM-DM270]

External Academic Service

- 2026 The European Conference on Computer Vision (ECCV) - Reviewer.
- 2026 Computer Vision and Image Understanding - Reviewer
- 2026 The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) - Emergency Reviewer.
- 2025 The Annual Conference on Neural Information Processing Systems (NeurIPS) - Reviewer.
- 2025 The International Conference on Computer Vision (ICCV) - Reviewer.
- 2025 IEEE Transactions on Multimedia - Reviewer.
- 2025 The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) - **Outstanding Reviewer.**
- 2024 The Annual Conference on Neural Information Processing Systems (NeurIPS) - Reviewer.

Most Relevant Publications

Google Scholar

- 2026 **Alex Costanzino**, P. Z. Ramirez, G. Lisanti, L. Di Stefano. *Spectral Gating via Damped Oscillations for Adaptive Implicit Neural Representations*. [Under Review]
- 2026 **Alex Costanzino**, P. Z. Ramirez, G. Lisanti, L. Di Stefano. *Modulate-and-Map: Crossmodal Feature Mapping with Cross-View Modulation for 3D Anomaly Detection*. [CVPR 2026 Findings]
- 2025 **Alex Costanzino**, W. Bayliss, J. Sock, M. Gorriz Blanch, D. Horak, I. Laptev, P. Torr, F. Pizzati. *Towards Reliable Identification of Diffusion-based Image Manipulations*. [NeurIPS 2025]
- 2025 **Alex Costanzino**, P. Z. Ramirez, L. Lella, M. Ragaglia, A. Oliva, G. Lisanti, L. Di Stefano. *SiM3D: Single-instance Multiview Multimodal and Multisetup 3D Anomaly Detection Benchmark*. [ICCV 2025]

- 2025 **Alex Costanzino**, P. Z. Ramirez, G. Lisanti, L. Di Stefano. *Learning to be a Transformer to Pinpoint Anomalies*. [IEEE Access]
- 2024 **Alex Costanzino**, P. Z. Ramirez, M. Del Moro, A. Aiezso, G. Lisanti, S. Salti, L. Di Stefano. *Test Time Training for Industrial Anomaly Segmentation*. [CVPR 2024 Workshops]
- 2024 **Alex Costanzino**, P. Z. Ramirez, G. Lisanti, L. Di Stefano. *Multimodal Industrial Anomaly Detection by Crossmodal Feature Mapping*. [CVPR 2024]
- 2023 **Alex Costanzino**, P. Z. Ramirez, M. Poggi, F. Tosi, S. Mattoccia, L. Di Stefano. *Learning Depth Estimation for Transparent and Mirror Surfaces*. [ICCV 2023]
- 2023 P. Z. Ramirez, **Alex Costanzino**, F. Tosi, M. Poggi, S. Salti, S. Mattoccia, L. Di Stefano. *Booster: a Benchmark for Depth from Images of Specular and Transparent Surfaces*. [IEEE TPAMI]

Organization of Workshops

- [W1] *NTIRE 2026: 11th New Trends in Image Restoration and Enhancement Workshop and Challenges*, **CVPR Workshops 2026** (Denver, Colorado). [<https://cvlab-unibo.github.io/booster-web/ntire26>]
- [W2] *TRICKY 2025: 3rd Transparent & Reflective objects In the wild Challenges*, **ICCV Workshops 2025** (Honolulu, Hawaii). [<https://sites.google.com/view/iccv25tricky>]
- [W3] *NTIRE 2025: 10th New Trends in Image Restoration and Enhancement Workshop and Challenges*, **CVPR Workshops 2025** (Nashville, Tennessee). [<https://cvlab-unibo.github.io/booster-web/ntire25>]
- [W4] *TRICKY 2024: 2nd Transparent & Reflective objects In the wild Challenges*, **ECCV Workshops 2024** (Milan, Italy). [<https://sites.google.com/view/eccv24-tricky-workshop/>]
- [W5] *NTIRE 2024: 9th New Trends in Image Restoration and Enhancement Workshop and Challenges*, **CVPR Workshops 2024** (Seattle, Washington). [<https://cvlab-unibo.github.io/booster-web/ntire24>]
- [W6] *NTIRE 2023: 8th New Trends in Image Restoration and Enhancement Workshop and Challenges*, **CVPR Workshops 2023** (Vancouver, Canada). [<https://cvlab-unibo.github.io/booster-web/ntire>]